Task 11: Session 11 (AIML) - Assignment Questions

Name: Mihir Sopan Patil

Domain: AIML

College & Branch Name: Zeal Polytechnic, Diploma In AIML

1. **Basic try-except**

   Write a program that takes two numbers as input from the user and prints their division.

   Use try-except to handle the case when the user enters non-numeric input (ValueError).

   Print a friendly message if an error occurs.

   Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The First Number:24
Enter The Second Number:j
Oops! That Doesn't Look Like A Valid Number. Please Enter Numeric Values.
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

   Program Code:

```python
try:
    num1 = float(input("Enter The First Number:"))
    num2 = float(input("Enter The Second Number:"))
    result = num1 / num2
    print("Result Of Division:", result)
except ValueError:
    print("Oops! That Doesn't Look Like A Valid Number. Please Enter Numeric
Values.")
```

2. Handling ZeroDivisionError

Write a program that divides two numbers entered by the user.

Use try-except to specifically catch ZeroDivisionError and print: "Error: Division by zero is not allowed."

Also handle ValueError for invalid input.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
rror.py
Enter The Numerator:64
Enter the Denominator:0
Error: Division By Zero Is Not Allowed.
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```
try:
    num1 = float(input("Enter The Numerator:"))
    num2 = float(input("Enter the Denominator:"))
    result = num1 / num2
    print("Result:", result)
except ZeroDivisionError:
    print("Error: Division By Zero Is Not Allowed.")
except ValueError:
    print("Error: Invalid Input. Please Enter Numeric Values Only.")
```

3. Multiple except Blocks

Create a program that performs division and also tries to convert a string to integer.

Handle both ValueError and ZeroDivisionError with separate except blocks and appropriate messages.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
.py
Enter The Numerator:5
Enter The Denominator:r
Error: Invalid Input! Could Not Convert The Given Value To A Number.
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```python
try:
    num1 = float(input("Enter The Numerator:"))
    num2 = float(input("Enter The Denominator:"))
    result = num1 / num2             # May Raise ZeroDivisionError

    text = input("Enter A String To Convert To Integer:")
    converted = int(text)            # May Raise ValueError
    print("Division Result:", result, "Converted Integer:", converted)

except ValueError:
    # Triggered When Input Cannot Be Converted To The Expected Numeric Type
    print("Error: Invalid Input! Could Not Convert The Given Value To A Number.")
except ZeroDivisionError:
    # Triggered When The Denominator Is Zero
    print("Error: You Cannot Divide A Number By Zero!")
```

4. Using else

Write a program that takes two numbers and divides them.

Use try, except, and else.

The else block should print "Division performed successfully!" only when no exception occurs.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Numerator:24
Enter The Denominator:3
Result: 8.0
Division Performed Successfully!
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```
try:
    num1 = float(input("Enter The Numerator:"))
    num2 = float(input("Enter The Denominator:"))
    result = num1 / num2
except ValueError:
    print("Error: Please Enter Valid Numeric Values.")
except ZeroDivisionError:
    print("Error: Division By Zero Is Not Allowed.")
else:
    # The else Block Runs Only When No Exception Was Raised In The try Block
    print("Result:", result)
    print("Division Performed Successfully!")
```

5. Using finally

Modify the division program to include a finally block that always prints: "Program execution completed."

Demonstrate that finally runs whether an exception occurs or not.

Add comments explaining the purpose of finally.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Numerator:24
Enter The Denominator:6
Result: 4.0
Program Execution Completed.
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```python
try:
    num1 = float(input("Enter The Numerator:"))
    num2 = float(input("Enter The Denominator:"))
    result = num1 / num2
    print("Result:", result)
except ValueError:
    print("Error: Invalid input. Please Enter Numbers Only.")
except ZeroDivisionError:
    print("Error: Division By Zero Is Not Allowed.")
finally:
    # The finally Block Always Executes Regardless Of Whether An Exception Occurred Or Not.
    print("Program Execution Completed.")
```

6. Combined try-except-else-finally

Write a complete division program using all four blocks (try, except, else, finally):

- Handle ValueError and ZeroDivisionError.
- Print success message in else.
- Always thank the user in finally.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
y.py
Enter The Numerator:82
Enter The Denominator:4
Result: 20.5
Division Performed Successfully!
Thank You For Using This Program!
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```python
try:
    num1 = float(input("Enter The Numerator:"))
    num2 = float(input("Enter The Denominator:"))
    result = num1 / num2
except ValueError:
    # Handles Non-Numeric Input
    print("Error: Please Enter Valid Numeric Values.")
except ZeroDivisionError:
    # Handles Division By Zero
    print("Error: Division By Zero Is Not Allowed.")
else:
    # Runs Only When The try Block Succeeds Without Any Exception
    print("Result:", result)
    print("Division Performed Successfully!")
finally:
    # Always Runs - Good Place To Thank The User Or Clean Up Resources
    print("Thank You For Using This Program!")
```

7. Practical - Age Input

Write a program that asks the user for their age.

- Convert the input to integer.
- If the age is negative, raise a ValueError with message "Age cannot be negative."
- Handle the exception gracefully and print a suitable message.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

Please Enter Your Age:-8
Error: Age Cannot Be Negative.
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```python
try:
    age = int(input("Please Enter Your Age:"))
    if age < 0:
    # Manually Raising A ValueError With A Descriptive Message
        raise ValueError("Age Cannot Be Negative.")
    print("Your Age Is:", age)
except ValueError as e:
        # As e Captures The Exception Object So We Can Print Its Message
    print("Error:", e)
```

8. Multiple Exceptions In One Block

Create a program that:

- Takes a number as input.
- Divides 100 by that number.
- Handle both ValueError and ZeroDivisionError in a single except block using a tuple.

Print a combined error message.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
_One_Block.py
Enter A Number To Divide 100 By:0
Error: Please Enter A Valid Non-Zero Number.(Invalid Input Or Division By Zero Detected.)
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```python
try:
    num = float(input("Enter A Number To Divide 100 By:"))
    result = 100 / num
    print("Result:", result)
except (ValueError, ZeroDivisionError):
    # A Single except Block Handling Two Exception Types Via A Tuple
    print("Error: Please Enter A Valid Non-Zero Number."
          "(Invalid Input Or Division By Zero Detected.)")
```

9. Debugging Exception Code

The following code has issues. Correct it and add comments explaining the fixes:

```
try
    num = int(input("Enter a number: "))
    result = 100 / num
    print("Result:", result)
except ValueError
    print("Please enter a valid number")
except:
print("Some error occurred")
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
de.py
Enter A Number:6
Result: 16.666666666666668
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```
try:                                           # Fix 1: Added ':'
    num = int(input("Enter A Number:"))
    result = 100 / num
    print("Result:", result)
except ValueError:                             # Fix 2: Added ':'
    print("Please Enter A Valid Number")
except ZeroDivisionError:                      # Fix 3: Added specific handler
    print("Error: Cannot Divide By Zero!")
except Exception as e:                         # Generic Catch-Allj
    print("Some Error Occurred:", e)
```

10. Mini Project - Exception Handling

Create a **Simple Calculator** with exception handling:

The program should run in a loop and show this menu:

a) Addition
b) Subtraction
c) Multiplication
d) Division
e) Exit

- Take user choice and two numbers.
- Perform the operation inside try block.
- Handle ValueError (invalid input) and ZeroDivisionError (for division).
- Use finally to print "Operation attempted." after every calculation.
- Continue until user chooses to exit.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

***** Calculator Menu *****
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter Your Choice (1-5): 1
Enter The First Number:g
Error: Invalid Input! Please Enter Numeric Values.
Operation Attempted.

***** Calculator Menu *****
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter Your Choice (1-5): 4
Enter The First Number:3
Enter The Second Number:0
Error: Division By Zero Is Not Allowed!
Operation Attempted.

***** Calculator Menu *****
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter Your Choice (1-5): 5
Exiting The Calculator. Goodbye!
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

Program Code:

```
while True:
    # Display menu
    print("\n***** Calculator Menu *****")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. Exit")

    choice = input("Enter Your Choice (1-5): ")
```

```python
# Exit Condition Checked Before Entering Try Block
if choice == "5":
    print("Exiting The Calculator. Goodbye!")
    break

try:
    num1 = float(input("Enter The First Number:"))
    num2 = float(input("Enter The Second Number:"))

    if choice == "1":
        result = num1 + num2
        print(f"Result: {num1} + {num2} = {result}")
    elif choice == "2":
        result = num1 - num2
        print(f"Result: {num1} - {num2} = {result}")
    elif choice == "3":
        result = num1 * num2
        print(f"Result: {num1} × {num2} = {result}")
    elif choice == "4":
        result = num1 / num2           # May Raise ZeroDivisionError
        print(f"Result: {num1} / {num2} = {result}")
    else:
        print("Invalid Choice! Please Select A Number Between 1 & 5.")

except ValueError:
    # Triggered When The User Enters Non-Numeric Input For A Number
    print("Error: Invalid Input! Please Enter Numeric Values.")
except ZeroDivisionError:
    # Triggered Only For Division When The Denominator Is Zero
    print("Error: Division By Zero Is Not Allowed!")
finally:
    # Runs After Every Iteration Regardless Of Success Or Error
    print("Operation Attempted.")
```